Integration as the limit of a sum

Exam code: 9MA0

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Written by: Paul
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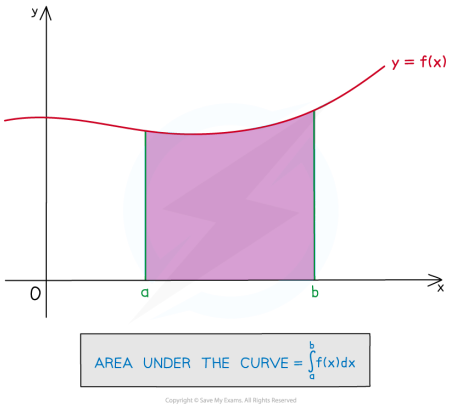
Integration as the limit of a sum

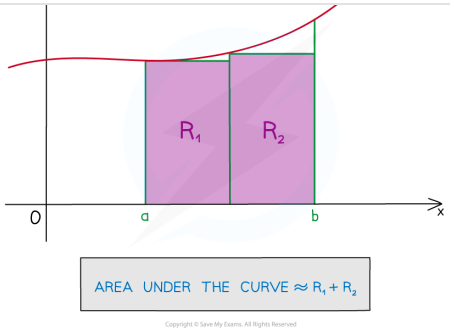
How do I represent the area under a curve using rectangles?

- Definite integration allows us to find the area under a curve

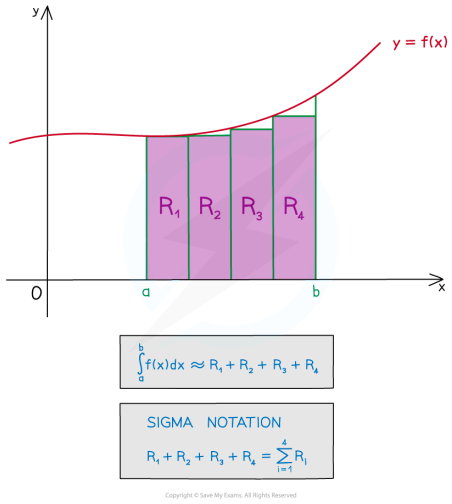


- An **estimate** for the area under the curve is the sum of the rectangular areas

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

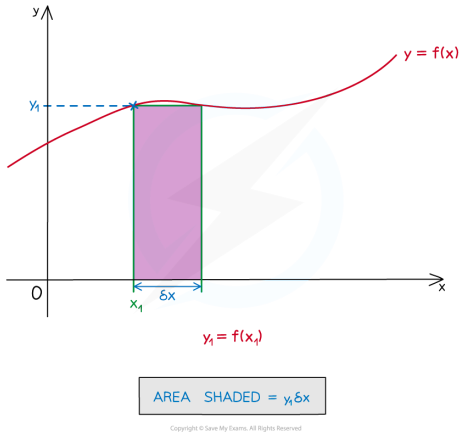


- If the number of rectangles increases and their width decreases, the estimate is more accurate

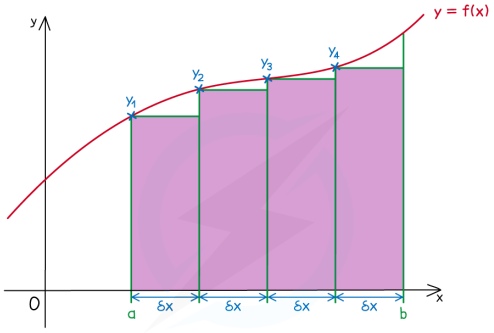


- The sum of the rectangle areas will have a limit, however small they get
  - The sum will become closer and closer to the area under the curve
  - This is called the **limit of the sum**

Why is integration seen as the limit of a sum?



- The height (length) will be the **y**-coordinate at $\mathbf{x_1}$ – ie $\mathbf{f(x_1)}$ (rather than $\mathbf{f(x_1+\delta x)}$)
- If we use four of these small rectangles between a and b we get



$$\text{SUM OF RECTANGLE AREAS} = \delta x y_1 + \delta x y_2 + \delta x y_3 + \delta x y_4$$
$$= \delta x(y_1 + y_2 + y_3 + y_4)$$
$$= \sum_{i=1}^{4} y_i \delta x$$

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- As more rectangles are used ...
  - ... **δx** gets smaller and smaller, ie $\boldsymbol{\delta x \to 0}$
  - ... **n**, the number of rectangles, gets bigger and bigger, ie $\mathbf{n \to \infty}$
  - ... the sum of the area of the rectangles becomes closer to the area under the curve

- This is the meaning of **integration as the limit of a sum**

How do I convert the limit of a sum into a definite integral?

- STEP 1 Recognise the notation
- STEP 2 Convert to a definite integral
- STEP 3 Find the value of the integral

STEP 1 RECOGNISE THE NOTATION

$f(x) = 3x^2 - 4$

UPPER LIMIT, $b = 5$

LOWER LIMIT, $a = 2$

STEP 2 CONVERT TO AN INTEGRAL

$$\lim_{\delta x \to 0} \sum_{x=2}^{5} (3x^2 - 4)\delta x = \int_2^5 (3x^2 - 4)\,dx$$

STEP 3 FIND THE VALUE OF THE INTEGRAL

$$\begin{aligned}\int_2^5 (3x^2 - 4)\,dx &= \left[x^3 - 4x\right]_2^5 \\ &= (5^3 - 4 \times 5) - (2^3 - 4 \times 2) \\ &= 105 - 0 \\ &= 105\end{aligned}$$

UNITS MAY BE NEEDED IF QUESTION HAD REFERRED TO AREA

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